Digitizing Loan Approvals for Financial Inclusion

Using Machine Learning to Power Smarter Credit for Kenyan MFIs & Digital Lenders

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Kenya's Financial Landscape Challenge

MFIs: Traditional Lifeline

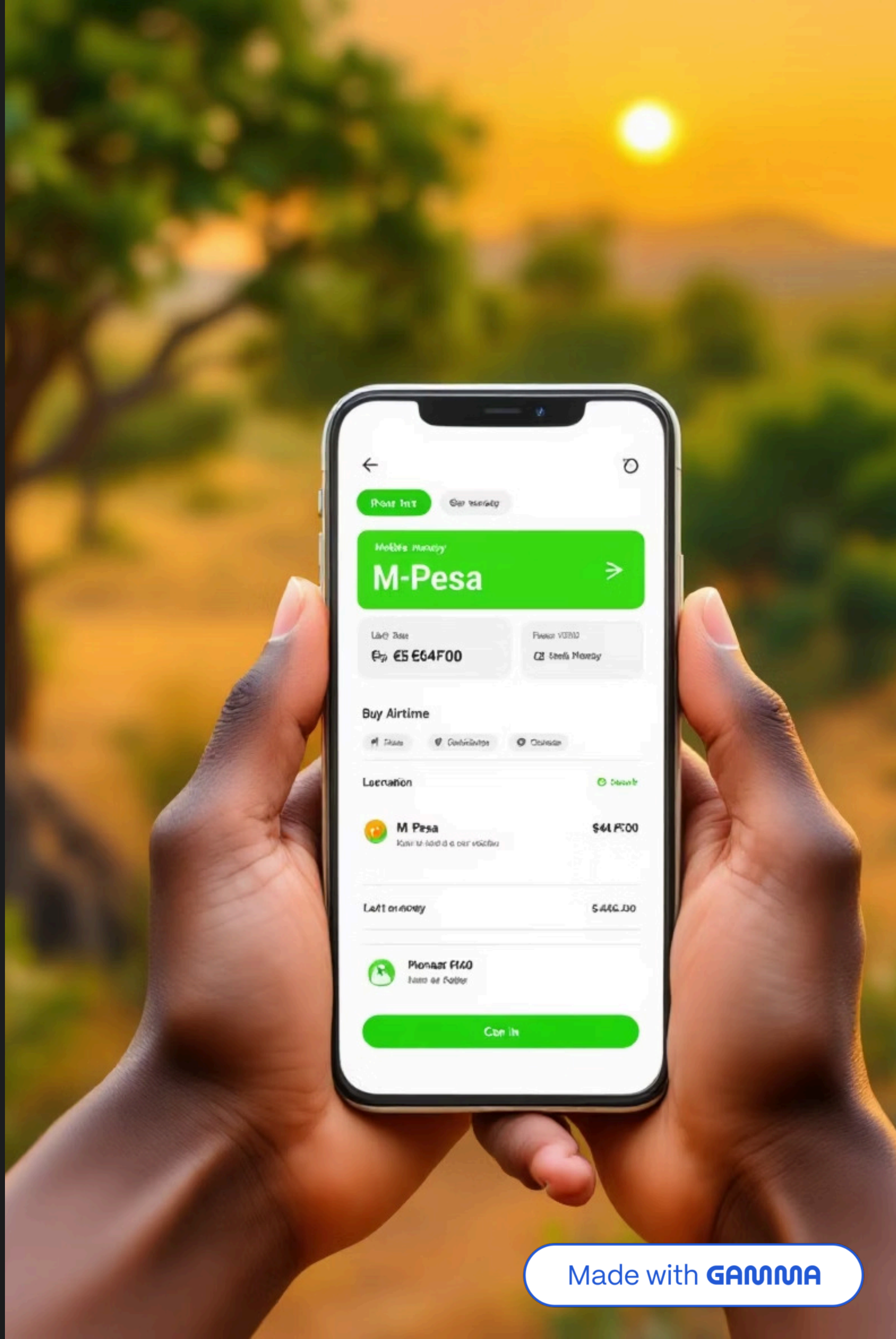
Serve underserved communities, women, and small businesses but face slow manual approvals and limited digitization

Digital Lenders: Tech-Powered

Instant AI-driven approvals via M-Pesa integration, but struggle with data privacy and rural access

Our Solution

Bridge the gap with machine learning models for faster, fairer, and more efficient credit decisions



Stakeholders & Value Proposition

MicroFinance Institutions (MFI's)

Must digitize to cut costs, reduce delays, and stay competitive in evolving market e.g Faulu, Rafiki

Digital Lenders

Provide instant credit but face privacy, security, and inclusion gaps e.g Branch, Tala

Applicants

Women, youth, rural entrepreneurs, and small & medium enterprises (SME's) need faster and fairer credit access

Regulators

Demand transparency, responsible lending, and sustainable growth e.g Central Bank of Kenya (CBK)

Value Proposition

- Automated approvals that reduce bureaucracy
- Fairer, data-driven credit scoring for underserved borrowers
- Stronger compliance and consumer protection for regulators

Objective

Deliver a supervised ML model for automated, fair, and efficient loan approvals, supporting Kenya's transition to a more inclusive and resilient financial sector

Data Overview

Dataset Foundation

Loan applications sourced from the **Hugging Face community platform**, containing financial, demographic, and outcome details

Target variable: loan_status (Approved vs. Not Approved)

Key Features

- **Loan Features:** loan_amnt (amount), term (duration), int_rate (interest rate)

Captures the core structure of a loan; critical for assessing affordability, risk exposure, and repayment likelihood

- **Credit Indicators:** grade / sub_grade

Provides proxy measures of borrower creditworthiness, directly influencing default probability

- **Applicant Information:** emp_length, home_ownership, annual_inc (income)

Reflects borrower stability and capacity to repay, key for fair and data-driven approvals

- **Purpose of Loan:** purpose (e.g., debt consolidation, credit card, home improvement)

Certain purposes correlate strongly with risk patterns, helping lenders adjust approval and pricing strategies

- **Outcome:** loan_status (target)

Defines the success of the prediction task; the benchmark against which model performance is evaluated

Data Challenges

- **Data Accessibility and Quality**

Finding a reliable, publicly available dataset with Kenyan-relevant loan data was difficult

Some datasets lacked complete borrower information, requiring cleaning, formatting, and standardization before modeling

- **Class Imbalance in Loan Outcomes**

The dataset had far more approved loans than defaults, which risked biasing the model.

We solved this using **SMOTE** to balance the classes and improve prediction fairness

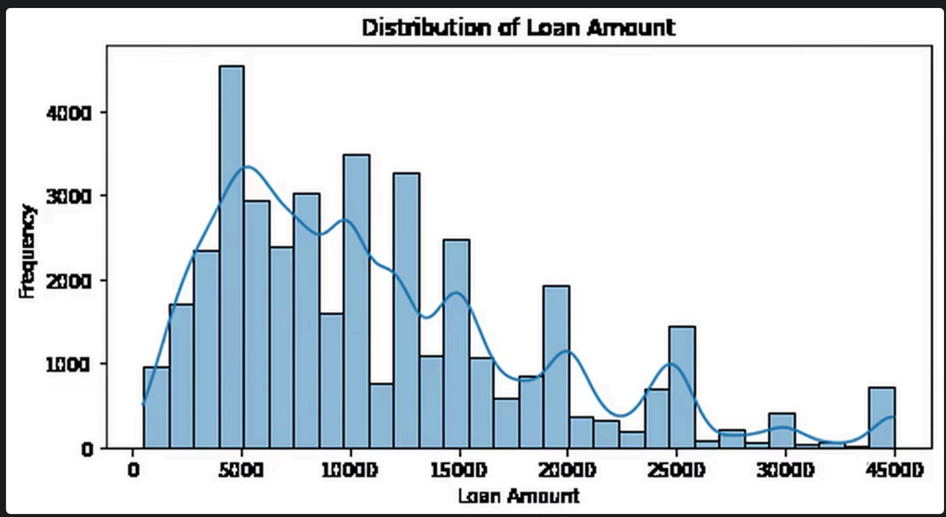
Data Insights from EDA

Loan Amounts

As we can see, most borrowers requested modest loans around 5,000, typical of microfinance lending patterns

Only a small portion took larger loans above 25,000, appearing as outliers

This pattern helps lenders understand borrower demand and set appropriate loan limits for different segments

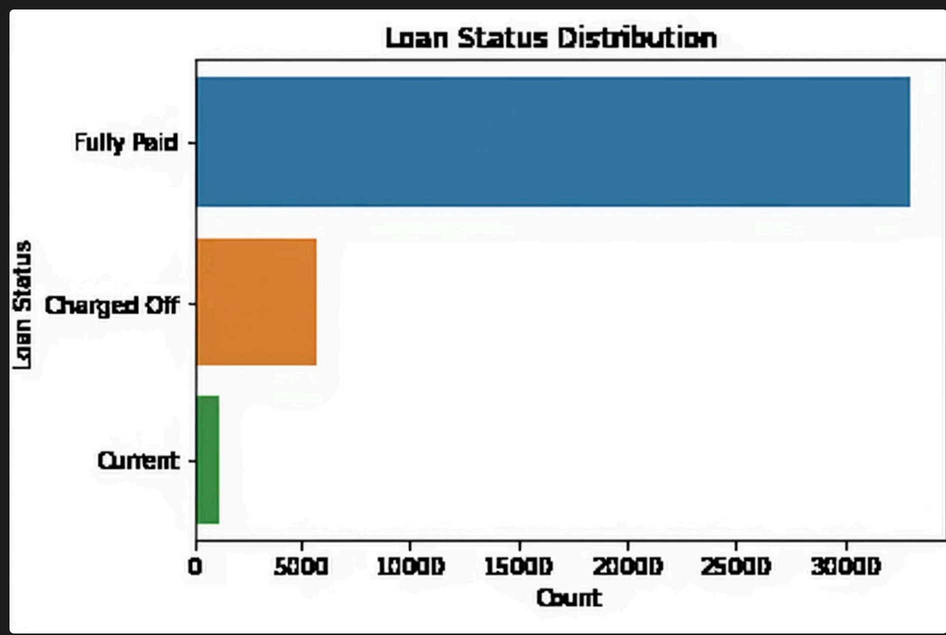


Loan Outcomes

The majority of borrowers fully repaid their loans, while defaults were rare but carried significant financial impact

Only a small portion of loans ended in default, highlighting the imbalance in the dataset

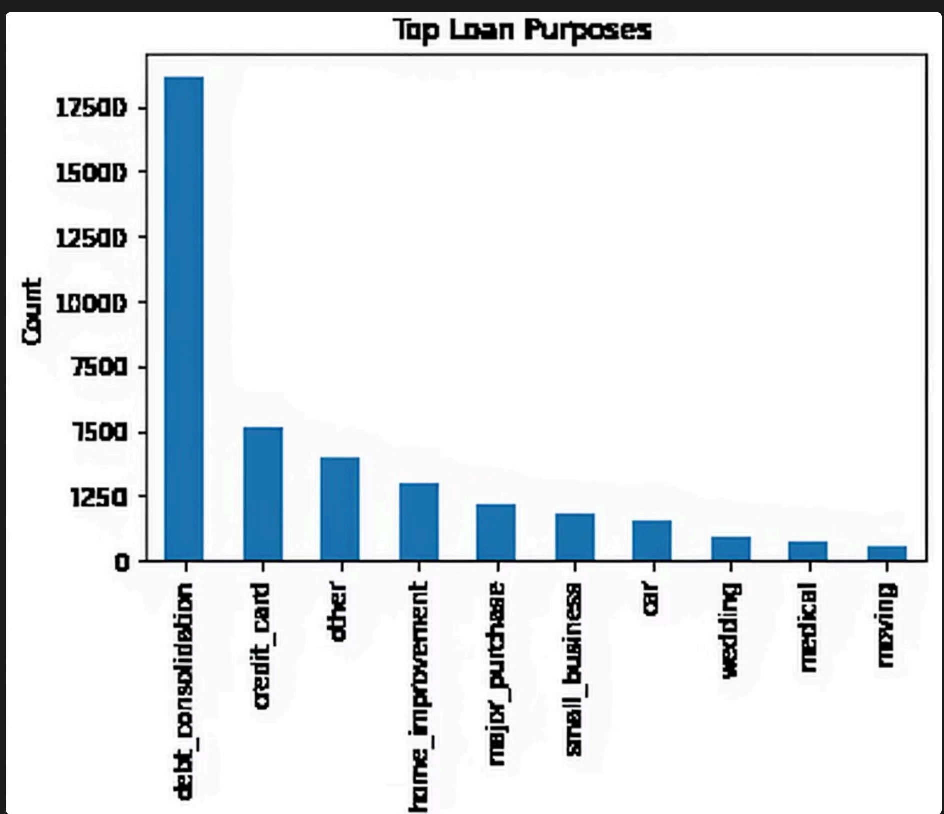
This confirms the need to focus on PR-AUC, which better evaluates model performance on these rare but important default cases



Loan Purposes

Debt consolidation is the most common reason for borrowing, followed by credit card refinancing and other personal uses, while moving is the least common

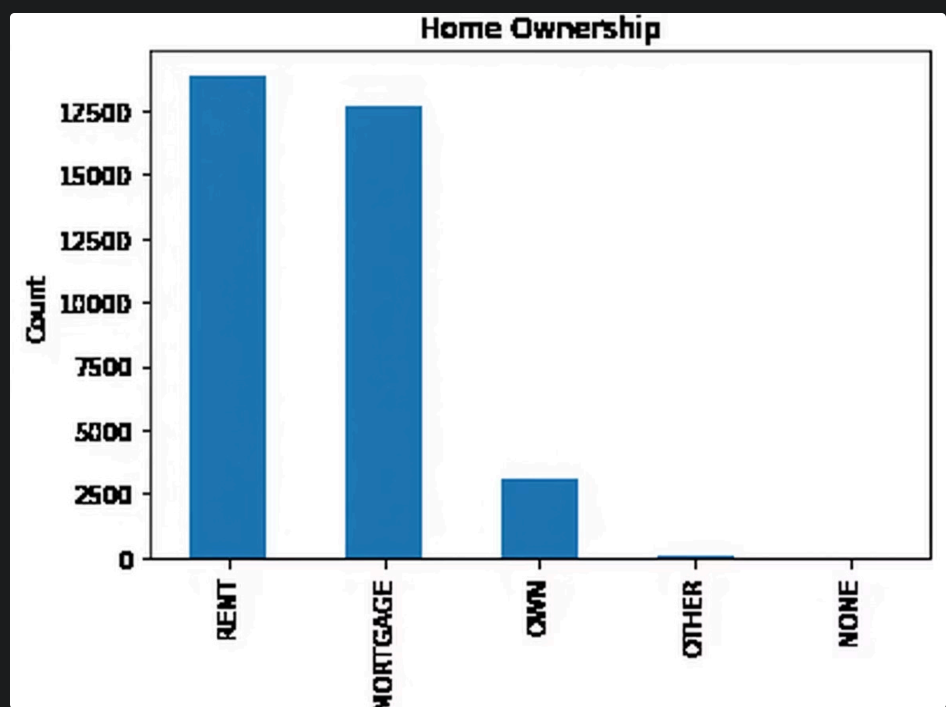
This pattern indicates that certain loan purposes carry higher risk, helping lenders tailor approval and pricing strategies



Home Ownership

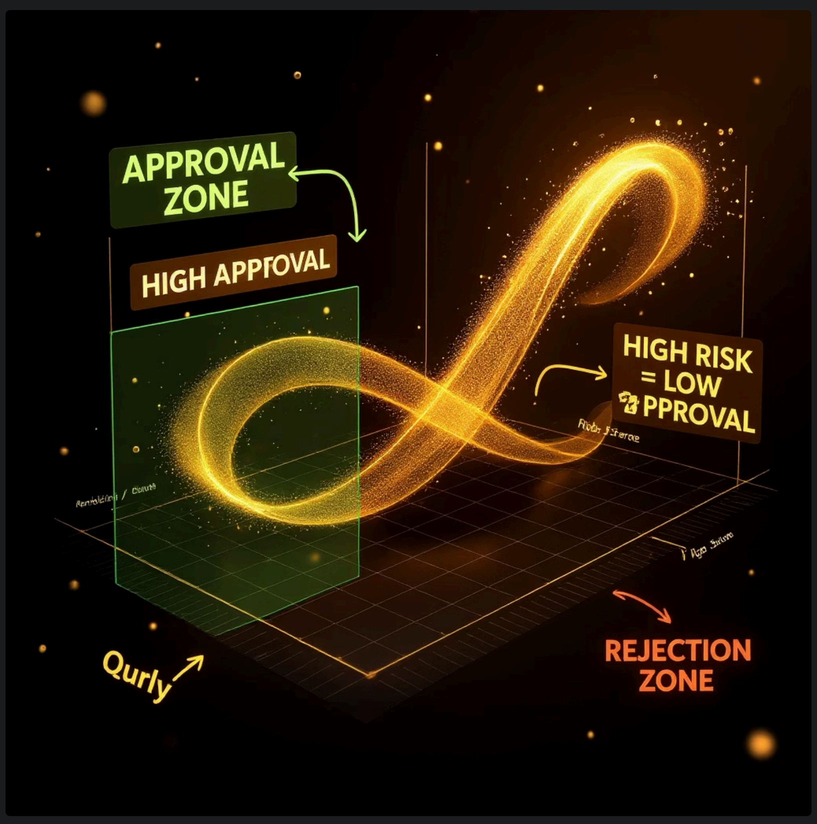
Most borrowers either rent or have mortgages, while those who fully own homes borrow less frequently

This pattern suggests that ownership type is a strong indicator of repayment ability and overall borrower stability



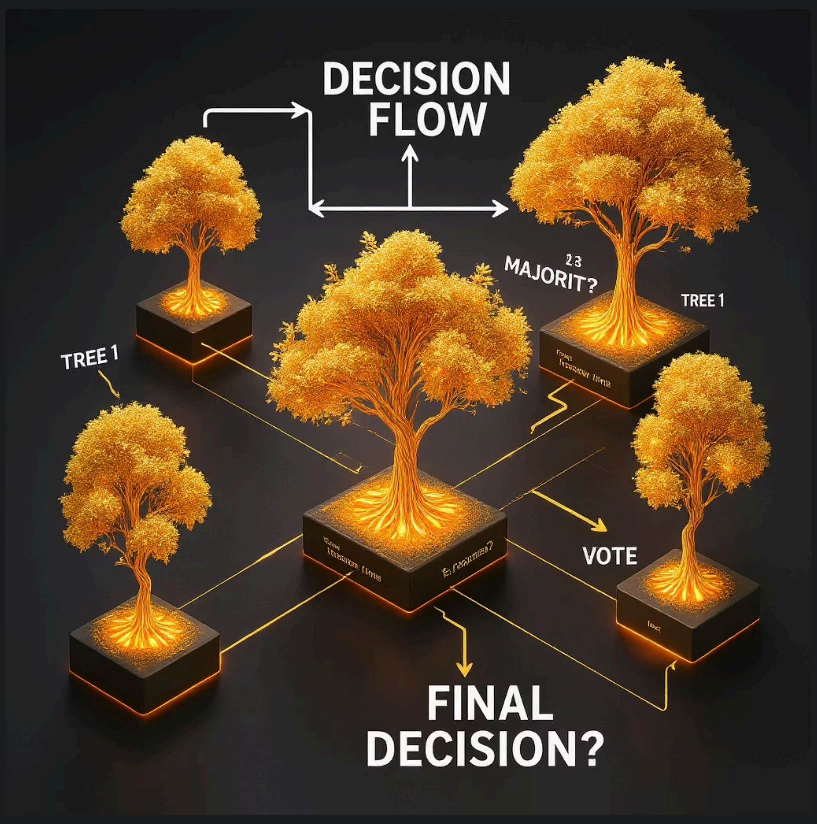
Modeling Approach

We tested four machine learning models to automate loan approvals, each bringing unique strengths:



Logistic Regression

This model served as our statistical baseline for predicting loan approval probabilities. We trained it on standardized numeric features and label-encoded categorical variables using balanced data after applying SMOTE. It was straightforward to implement, easy to interpret, and provided a reliable benchmark for evaluating the performance of more advanced models.



Random Forest

This model was trained using multiple decision trees built on bootstrapped samples of the training data. It averages their predictions to reduce bias and variance, allowing it to capture complex non-linear relationships between features such as loan amount, grade, and income. It delivered higher accuracy than Logistic Regression but was less interpretable.

XGBoost

This gradient boosting model was trained by building trees sequentially, where each new tree corrected the errors of the previous ones. It efficiently handled missing values and outliers while optimizing for precision and recall using imbalanced loan data. XGBoost delivered the strongest overall performance, achieving the best precision-recall balance among all models.

Stacking Ensemble

This model combined predictions from the Random Forest and XGBoost models, using XGBoost as the meta-learner to refine final predictions. It produced a slight improvement in metrics compared to individual models but introduced additional complexity. The marginal performance gain did not translate into significant business value.

Key Performance Metrics

PR-AUC

Precision-Recall Area Under Curve focuses on correctly identifying approved loans. Crucial for imbalanced data and aligns with the business goal of maximizing fair approvals while minimizing risk

Recall

Measures actual approved loans correctly identified. High recall means fewer missed opportunities for eligible applicants

Precision

Measures how many predicted approvals are actually correct. High precision reduces the risk of lending to risky applicants

F1 Score

Balances precision and recall, providing single measure of overall model quality

Model Performance Results

We compared four models on the holdout test set using key metrics:

Model	PR-AUC	Precision	Recall	F1 Score	Interpretation
Logistic Regression	0.897	0.889	0.573	0.697	With a PR-AUC of **0.897**, the model was simple and easy to interpret but missed several defaults due to lower recall **0.573**, increasing potential risk of losses.
Random Forest	0.913	0.864	0.832	0.847	Achieved a PR-AUC of **0.913** with improved recall **0.832** and balanced precision **0.864** However, its predictions were slightly inconsistent across runs
XGBoost	**0.915**	**0.863**	**0.871**	**0.867**	Delivered the highest PR-AUC of **0.915** and strong recall **0.871** with balanced precision **0.863**, making it the most reliable model for minimizing default risk while approving more good borrowers
Stacking Ensemble	0.911	0.848	0.917	0.881	Recorded a PR-AUC of **0.911** with the highest recall **0.917**, but its added complexity and minimal improvement over XGBoost made it less practical for deployment

Why XGBoost Wins

Best Performance

Highest PR-AUC (0.915) → most suitable metric for rare defaults in loan data

Balanced Risk

High recall (0.871) → fewer deserving applicants wrongly rejected, allowing more eligible borrowers to access credit responsibly

High precision (0.863) → fewer risky loans approved, reducing default risk and improving overall portfolio quality

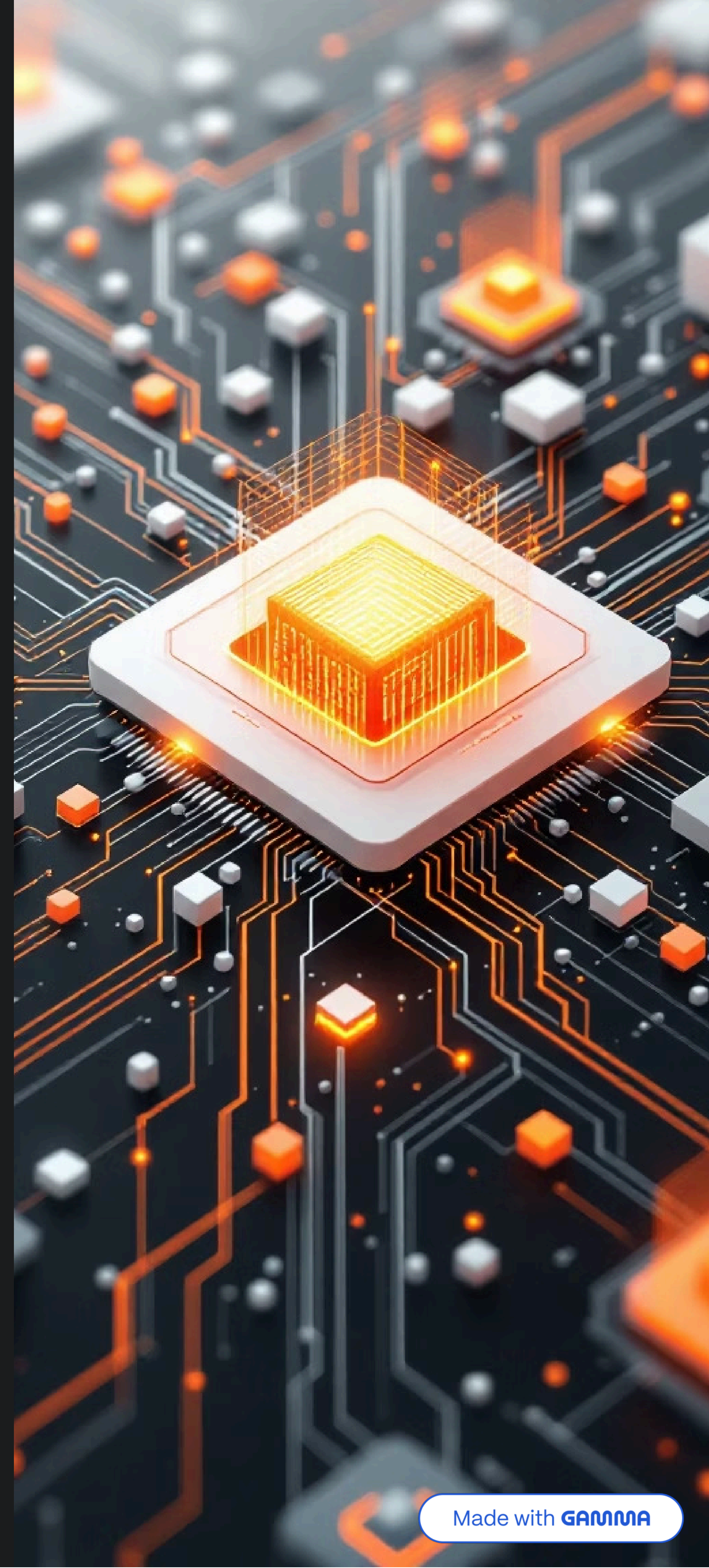
Imbalanced Data Handling

Focuses on learning from rare default cases, maintaining high precision and recall even when approved loans greatly outnumber defaults—making it more dependable than models that rely on accuracy alone

Business Ready

Robust to outliers, handles missing data, fast processing, & provides clear feature importances

XGBoost outperformed all models making it the most effective and practical model for fair, efficient, and scalable loan approvals delivering real financial inclusion



Integrating Sentiment Analysis with XGBoost

To further enhance our loan approval process, we are integrating sentiment analysis, providing a deeper understanding of applicant intent and tone

Overview

Borrower descriptions are analyzed using VADER sentiment analysis to capture tone and intent, adding a sentiment score to the XGBoost model for more informed decisions

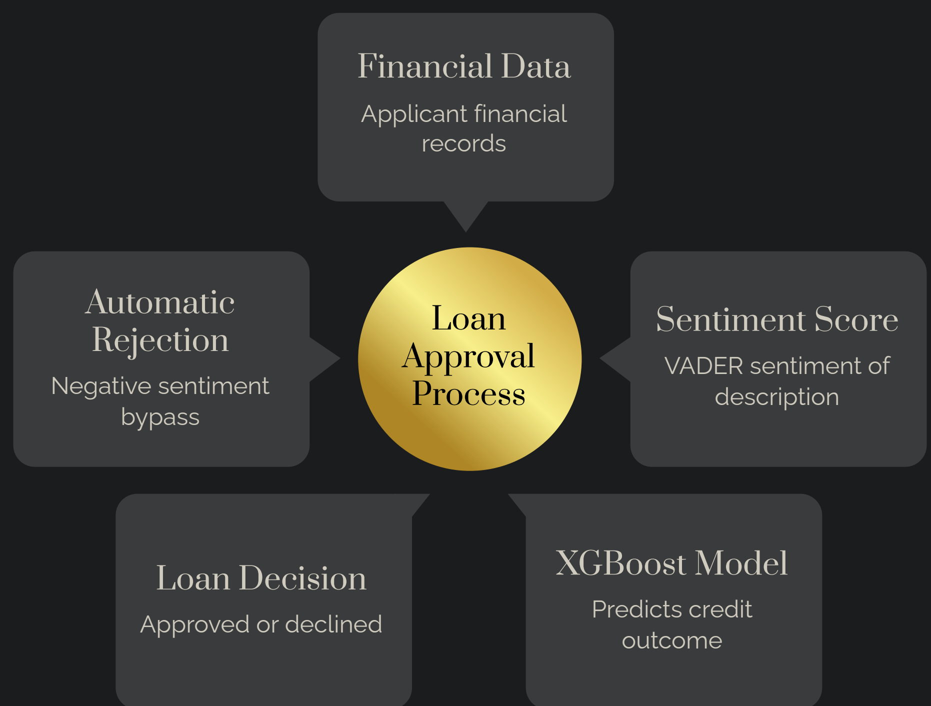
Sentiment-Based Loan Rule

If a borrower's sentiment score is too negative, the application is automatically flagged or rejected. Otherwise, the model evaluates the loan as usual

Business Impact

Adding sentiment improves risk detection and ensures fair, responsible lending decisions by considering both quantitative data and qualitative borrower behavior

This flow diagram illustrates how sentiment analysis is integrated into our enhanced loan approval process:



Conclusion

Business Value

- Identifies ~87% of eligible applicants → fewer deserving rejections
- 86% precision → lowers lender's exposure to bad loans
- Strong PR-AUC → consistent performance across thresholds

Operational Recommendations

- Deploy in decision pipeline with feature transformations
- Start conservative, A/B test for trade-offs
- Monitor drift & fairness monthly
- Retrain regularly with governance checklist

Next Steps

- Retrain regularly & adopt a model governance checklist
- Integrate additional data sources (e.g., mobile transactions or credit bureau data) to enhance prediction accuracy and inclusiveness

Empowering financial inclusion through faster, fairer, and smarter lending 🙏

Thank you

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💡 *"Together, we're shaping the future of inclusive finance in Kenya — one intelligent, data-driven loan decision at a time."*